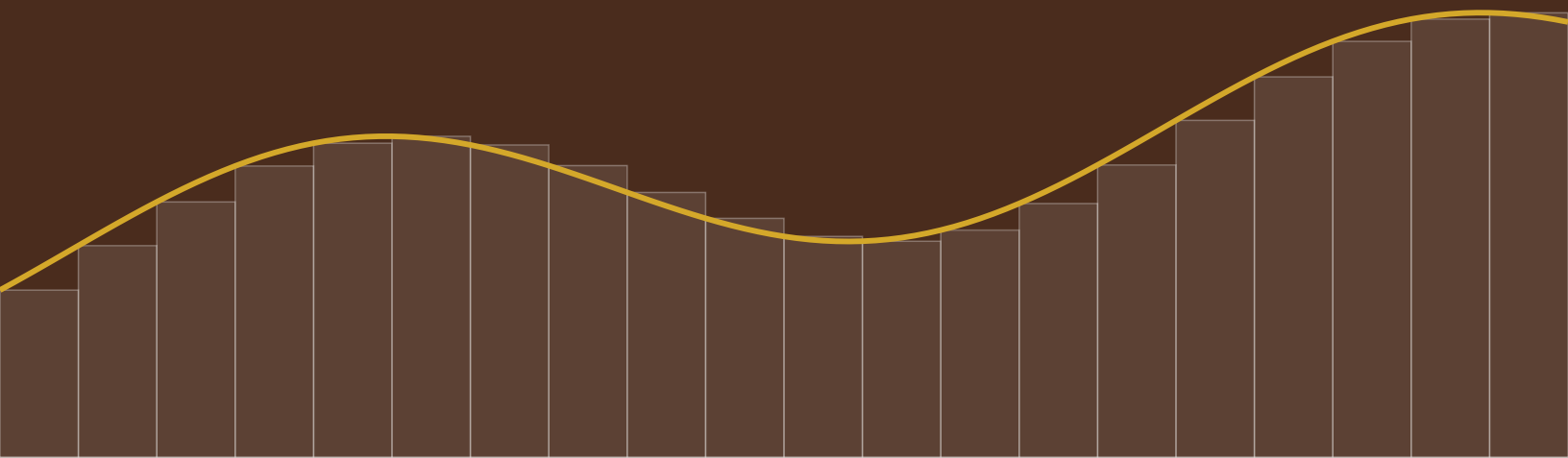




CARMEL CATHOLIC HIGH SCHOOL

AP Calculus BC

Course Notes

Based on Hass, Joel, et al. *Thomas' Calculus: Early Transcendentals*. 15th ed., Global ed., Pearson Education Limited, 2024.

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CHAPTER 1

Functions

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NOTE This chapter is pretty much review from Precalculus. I didn't even study this and took the quiz and got an A, so I think this chapter is supplemental. But it's in the textbook, and the following information relies on this.

1.1 Functions and Their Graphs

Functions; Domain and Range

Graphs of Functions

Representing a Function Numerically

The Vertical Line Test for a Function

Piecewise-Defined Functions

Increasing and Decreasing Functions

Even Functions and Odd Functions: Symmetry

Common Functions

1.2 Combining Functions; Shifting and Scaling Graphs

Sums, Differences, Products, and Quotients

Composing Functions

Shifting a Graph of a Function

Scaling and Reflecting a Graph of a Function

1.3 Trigonometric Functions

Angles

The Six Basic Trigonometric Functions

Periodicity and Graphs of the Trigonometric Functions

Trigonometric Identities

The Law of Cosines

Transformations of Trigonometric Graphs

Two Special Inequalities

1.4 Exponential Functions

Exponential Behavior

The Natural Exponential Function e^x

Exponential Growth and Decay

1.5 Inverse Functions and Logarithms

One-to-One Functions

Inverse Functions

Finding Inverses

Logarithmic Functions

Properties of Logarithms

Applications

Inverse Trigonometric Functions

The Arcsine and Arccosine Functions

Identities Involving Arcsine and Arccosine

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Limits and Continuity

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2.1 Rates of Change and Tangent Lines to Curves

Average and Instantaneous Speed

Defining the Slope of a Curve

Average Rates of Change and Secant Lines

Rates of Change and Tangent Lines

2.2 Limit of a Function and Limit Laws

An Informal Description of the Limit of a Function

Limits of Function Values

The Limit Laws

Eliminating Common Factors from Zero Denominators

Evaluating Limits of Polynomials and Rational Functions

Using Calculators and Computers to Estimate Limits

The Sandwich Theorem

2.4 One-Sided Limits

Approaching a Limit from One Side

Limits at Endpoints of an Interval

Precise Definitions of One-Sided Limits

Limits Involving $(\sin \theta)/\theta$

2.5 Limits Involving Infinity; Asymptotes of Graphs

Finite Limits as $x \rightarrow \pm\infty$

Limits at Infinity of Rational Functions

Horizontal Asymptotes

Oblique Asymptotes

Infinite Limits

Precise Definitions of Infinite Limits

Vertical Asymptotes

Dominant Terms

2.6 Continuity

Continuity at a Point

Continuous Functions

Inverse Functions and Continuity

Continuity of Compositions of Functions

Intermediate Value Theorem for Continuous Functions

Continuous Extension to a Point

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Derivatives

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3.1 Tangent Lines and the Derivative at a Point

Finding a Tangent Line to the Graph of a Function

Rates of Change: Derivative at a Point

3.2 The Derivative as a Function

Calculating Derivatives from the Definition

Notation

Graphing the Derivative

Differentiability on an Interval; One-Sided Derivatives

When Does a Function Not Have a Derivative at a Point?

Differentiable Functions Are Continuous

3.3 Differentiation Rules

Powers, Multiples, Sums, and Differences

Derivatives of Exponential Functions

Products and Quotients

Second- and Higher-Order Derivatives

3.4 The Derivative as a Rate of Change

Instantaneous Rates of Change

Motion Along a Line: Displacement, Velocity, Speed, Acceleration, and Jerk

Derivatives in Economics and Biology

Sensitivity to Change

3.5 Derivatives of Trigonometric Functions

Derivative of the Sine Function

Derivative of the Cosine Function

Simple Harmonic Motion

Derivatives of the Other Basic Trigonometric Functions

3.6 The Chain Rule

Derivative of a Composite Function

“Outside-Inside” Rule

Repeated Use of the Chain Rule

The Chain Rule with Powers of a Function

3.7 Implicit Differentiation

Implicitly Defined Functions

Derivatives of Higher Order

Lenses, Tangent Lines, and Normal Lines

3.8 Derivatives of Inverse Functions and Logarithms

Derivatives of Inverses of Differentiable Functions

Derivative of the Natural Logarithm Function

The Derivatives of a^x and $\log_a x$

Logarithmic Differentiation

Irrational Exponents and the Power Rule (General Version)

The Number e Expressed as a Limit

3.9 Inverse Trigonometric Functions

Inverses of $\tan x$, $\cot x$, $\sec x$, and $\csc x$

The Derivative of $y = \arcsin u$

The Derivative of $y = \arctan u$

The Derivative of $y = \operatorname{arcsec} u$

Derivatives of the Other Three Inverse Trigonometric Functions

3.10 Related Rates

Related Rates Equations

3.11 Linearization and Differentials

Linearization

Differentials

Estimating with Differentials

Error in Differential Approximation

Proof of the Chain Rule

Sensitivity to Change

Converting Mass to Energy

Applications of Derivatives

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4.1 Extreme Values of Functions on Closed Intervals

Local (Relative) Extreme Values

Finding Extrema

4.2 The Mean Value Theorem

Rolle's Theorem

The Mean Value Theorem

A Physical Interpretation

Mathematical Consequences

Finding Velocity and Position from Acceleration

4.3 Monotonic Functions and the First Derivative Test

Increasing Functions and Decreasing Functions

First Derivative Test for Local Extrema

4.4 Concavity and Curve Sketching

Concavity

Points of Inflection

Second Derivative Test for Local Extrema

Graphical Behavior of Functions from Derivatives

4.5 Indeterminate Forms and L'Hôpital's Rule

Indeterminate Form $0/0$

Indeterminate Forms ∞/∞ , $\infty \cdot 0$, $\infty - \infty$

Indeterminate Powers

Proof of L'Hôpital's Rule

4.6 Applied Optimization

Examples from Mathematics and Physics

Examples from Economics

4.8 Antiderivatives

Finding Antiderivatives

Initial Value Problems and Differential Equations

Antiderivatives and Motion

Indefinite Integrals

CHAPTER 5

Integrals

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5.1 Area and Estimating with Finite Sums

Area

Distance Traveled

Displacement Versus Distance Traveled

Average Value of a Nonnegative Continuous Function

5.2 Sigma Notation and Limits of Finite Sums

Finite Sums and Sigma Notation

Limits of Finite Sums

Riemann Sums

5.3 The Definite Integral

Definition of the Definite Integral

Integrable and Nonintegrable Functions

The Midpoint Rule

Properties of Definite Integrals

Area Under the Graph of a Nonnegative Function

Average Value of a Continuous Function Revisited

5.4 The Fundamental Theorem of Calculus

Mean Value Theorem for Definite Integrals

Fundamental Theorem, Part 1

Fundamental Theorem, Part 2 (The Evaluation Theorem)

The Integral of a Rate

The Relationship Between Integration and Differentiation

Total Area

5.5 Indefinite Integrals and the Substitution Method

Substitution: Running the Chain Rule Backwards

Trying Different Substitutions

5.6 Definite Integral Substitutions and the Area Between Curves

The Substitution Formula

Definite Integrals of Symmetric Functions

Areas Between Curves

Integration with Respect to y

Applications of Definite Integrals

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6.1 Volumes Using Cross-Sections

NOTE Disk and washer methods, plus solids with known cross-sections. The AP exam does not require cylindrical shells (6.2).

Slicing by Parallel Planes

Solids of Revolution: The Disk Method

Solids of Revolution: The Washer Method

6.3 Arc Length

Length of a Curve $y = f(x)$

Dealing with Discontinuities in dy/dx

The Differential Formula for Arc Length

Integrals and Transcendental Functions

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7.2 Exponential Change and Separable Differential Equations

NOTE Separable differential equations and exponential growth and decay. Logistic growth is in 16.4.

Exponential Change

Separable Differential Equations

Unlimited Population Growth

Radioactivity

Heat Transfer: Newton's Law of Cooling

CHAPTER 8

Techniques of Integration

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8.1 Using Basic Integration Formulas

8.2 Integration by Parts

Product Rule in Integral Form

Evaluating Definite Integrals by Parts

8.5 Integration of Rational Functions by Partial Fractions

NOTE AP only tests partial fractions with distinct linear factors in the denominator.

General Description of the Method

Determining Coefficients by Differentiating

8.7 Numerical Integration

NOTE The Trapezoidal Rule is tested; Simpson's Rule is not.

Approximating Integrals with the Midpoint Rule

Trapezoidal Approximations

Simpson's Rule: Approximations Using Parabolas

Error Analysis

8.8 Improper Integrals

Infinite Limits of Integration

The Integral $\int_1^\infty dx/x^p$

Integrands with Vertical Asymptotes

Improper Integrals with a CAS

Tests for Convergence and Divergence

CHAPTER 9

Infinite Sequences and Series



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9.1 Sequences

Representing Sequences

Convergence and Divergence

Calculating Limits of Sequences

Using L'Hôpital's Rule

Commonly Occurring Limits

Recursive Definitions

Bounded Monotonic Sequences

9.2 Infinite Series

Geometric Series

The n th-Term Test for a Divergent Series

Combining Series

Adding or Deleting Terms

Reindexing

9.3 The Integral Test

The Integral Test

Nondecreasing Partial Sums

Error Estimation

9.4 Comparison Tests

The Limit Comparison Test

9.5 Absolute Convergence; The Ratio and Root Tests

NOTE The Ratio Test is tested; the Root Test is not.

The Ratio Test

The Root Test

9.6 Alternating Series and Conditional Convergence

Conditional Convergence

Rearranging Series

Summary of Tests to Determine Convergence or Divergence

9.7 Power Series

Power Series and Convergence

The Radius of Convergence of a Power Series

Operations on Power Series

9.8 Taylor and Maclaurin Series

Series Representations

Taylor and Maclaurin Series

Taylor Polynomials

9.9 Convergence of Taylor Series

NOTE Emphasize the Lagrange Error Bound (Taylor's Theorem with remainder).

Estimating the Remainder

Using Taylor Series

A Proof of Taylor's Theorem

9.10 Applications of Taylor Series

The Binomial Series for Powers and Roots

Evaluating Nonelementary Integrals

Arctangents

Evaluating Indeterminate Forms

Euler's Identity

Parametric Equations and Polar Coordinates

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10.1 Parametrizations of Plane Curves

Parametric Equations

Cycloids

Brachistochrones and Tautochrones

10.2 Calculus with Parametric Curves

Tangent Lines and Areas

Length of a Parametrically Defined Curve

Length of a Curve $y = f(x)$

The Arc Length Differential

Areas of Surfaces of Revolution

10.3 Polar Coordinates

Definition of Polar Coordinates

Polar Equations and Graphs

Relating Polar and Cartesian Coordinates

10.4 Graphing Polar Coordinate Equations

Symmetry

Slope

Converting a Graph from the $r\theta$ -Plane to the xy -Plane

10.5 Areas and Lengths in Polar Coordinates

NOTE AP tests area in polar coordinates. Polar arc length is not tested.

Area in the Plane

Length of a Polar Curve

CHAPTER 12

Vector-Valued Functions and Motion in Space

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12.1 Curves in Space and Their Tangents

NOTE AP covers plane curves only. Treat vector-valued functions as $\langle x(t), y(t) \rangle$.

Limits and Continuity

Derivatives and Motion

Differentiation Rules

Vector Functions of Constant Length

12.2 Integrals of Vector Functions; Projectile Motion

NOTE Focus on recovering position from velocity and on displacement vs. total distance traveled.

Integrals of Vector Functions

The Vector and Parametric Equations for Ideal Projectile Motion

Projectile Motion with Wind Gusts

First-Order Differential Equations

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16.1 Solutions, Slope Fields, and Euler's Method

NOTE Slope fields and Euler's method.

General First-Order Differential Equations and Solutions

Slope Fields: Viewing Solution Curves

Euler's Method

16.4 Graphical Solutions of Autonomous Equations

NOTE Logistic growth: the carrying capacity and where the population grows fastest.

Equilibrium Values and Phase Lines

Stable and Unstable Equilibria

Newton's Law of Cooling

A Falling Body Encountering Resistance

The Logistic Model for Population Growth

The Logistic Equation in Neural Networks and Machine Learning